Retail Scan Performance Analysis

1. Project Overview

This project analyzes sales performance, Discount Impact Analysis and Customer Behavior using transactional data from 12,575 transactions across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and Discount Impact on sales to guide strategic business decisions.

2. Dataset Summary

- Rows: 12,575
- Columns: 11
- Key Features:
- Transaction Details: (Transaction ID, Transaction Date)
- Customer Information: (Customer ID, Location, Payment)
- Purchase Details: (Item, Category, price per unit, Quantity, Total Spent)
- Shopping behavior: (Discount Applied)
- Missing Data: 1213 values in Item column, 609 values in Price Per Unit,

  604 values in Quantity, 604 values in Total Spent

  4199 values in Discount Applied

3. Exploratory Data Analysis using Python

I started with data preparation and cleaning in Python

- **Data Loading:** Imported the dataset using Pandas.

- **Initial Exploration:** Used df .info() to check the structure and .describe() for summary statistics.

	Transaction ID	Customer ID	Category	Item	Price Per Unit	Quantity	Total Spent	Payment Method	Location	Transaction Date	Discount Applied
count	12575	12575	12575	11362	11966.000000	11971.000000	11971.000000	12575	12575	12575	8376
unique	12575	25	8	200	NaN	NaN	NaN	3	2	1114	2
top	TXN_6867343	CUST_05	Electric household essentials	Item_2_BEV	NaN	NaN	NaN	Cash	Online	2022-05-30	True
freq	1	544	1591	126	NaN	NaN	NaN	4310	6354	26	4219
mean	NaN	NaN	NaN	NaN	23.365912	5.536380	129.652577	NaN	NaN	NaN	NaN
std	NaN	NaN	NaN	NaN	10.743519	2.857883	94.750697	NaN	NaN	NaN	NaN
min	NaN	NaN	NaN	NaN	5.000000	1.000000	5.000000	NaN	NaN	NaN	NaN
25%	NaN	NaN	NaN	NaN	14.000000	3.000000	51.000000	NaN	NaN	NaN	NaN
50%	NaN	NaN	NaN	NaN	23.000000	6.000000	108.500000	NaN	NaN	NaN	NaN
75%	NaN	NaN	NaN	NaN	33.500000	8.000000	192.000000	NaN	NaN	NaN	NaN
max	NaN	NaN	NaN	NaN	41.000000	10.000000	410.000000	NaN	NaN	NaN	NaN

- **Missing Data Handling:** Checked for null values and imputed missing values in the Item with 'Unknown', Discount Applied with 'No', Price per unit using mean, Quantity using the median, Total Spent using median by grouping it with the product Category.

- **Column Standardization:** Renamed columns to **snake case** for better readability and documentation.

- **Feature Engineering:**
  - Created new Columns **transaction year, transaction day** and **transaction month** by splitting the **transaction date** column.

- **Data Consistency Check:** Verified consistency between price_per_unit, quantity, and total_spent; ensured calculated totals align correctly and corrected any mismatches.

- **Database Integration:** Connected Python script to MySQL and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis using SQL (Business Transactions)

I have Performed structured analysis in MySQL to answer key business questions:

1. **Revenue by Category –** Compared total revenue generated across the product categories.

category	Revenue
Butchers	216704
Electric household essentials	212101
Beverages	204822
Food	203722
Furniture	201943
Computers and electric accessories	199602.5
Patisserie	191213.5
Milk Products	186644

2. **Total Sales by Category:** Compared Total Sales generated by Category

category	sales
Food	8873
Furniture	8858
Beverages	8784
Milk Products	8765
Electric household essentials	8759
Computers and electric accessories	8758
Butchers	8566
Patisserie	8465

3. **Revenue and Sales by Location:** Compared Total Sales and Total Revenue based on the Location.

location	Revenue	sales
Online	822101.75	35416
In-store	794650.25	34412

4. **Revenue by years:** Generated the Total Revenue over time by using Transaction years.

years	Revenue
2024	546700.75
2022	532761.5
2023	510611.75
2025	26678

5. **Sales Performance of Categories by Location:** Generated each Category of sales based on the Location.

category	location	sales
Computers and electric accessories	Online	4721
Electric household essentials	Online	4518
Beverages	Online	4483
Furniture	In-store	4474
Food	In-store	4442
Food	Online	4431
Milk Products	In-store	4388
Furniture	Online	4384
Milk Products	Online	4377
Butchers	Online	4322
Beverages	In-store	4301
Patisserie	In-store	4285
Butchers	In-store	4244
Electric household essentials	In-store	4241
Patisserie	Online	4180
Computers and electric accessories	In-store	4037

6. **Contribution of Revenue by each Category:** Generated Contribution Percentage how each Category contributed to Total Revenue calculated as Product Revenue.

category	Product_Revenue	contribution_percentage
Butchers	216704	13.403663641671697
Electric household essentials	212101	13.118957020000593
Beverages	204822	12.66873336170297
Food	203722	12.600695715854998
Furniture	201943	12.490660286797233
Computers and electric accessories	199602.5	12.345894732154344
Patisserie	191213.5	11.827014903955584
Milk Products	186644	11.544380337862579

7. **Total Spent by Customer ID:** Compared to the Spending by customer ID how each customer spent.

customer_id	spending
CUST_05	69973.75
CUST_23	67973.25
CUST_13	67859.25
CUST_16	67819.75
CUST_10	65244.25
CUST_21	65243.75
CUST_15	65025.5
CUST_22	64601.75
CUST_20	64441.75
CUST_02	64342.75
CUST_19	64030.5
CUST_07	63871.25
CUST_04	63819.5
CUST_09	63677.5
CUST_12	63522.25
CUST_11	63092
CUST_03	62840.75
CUST_18	62565.25

8. **Average Spending of Customers based on location:** Compared Average Spending of Customers based on the Location.

location	spending
Online	129.3833412023922
In-store	127.73673846648448

9. **Customer Purchase by Transaction ID:** Compared the Customer IDs of Customers who Purchase frequently by Sales.

customer_id	purchase_frequently
CUST_05	544
CUST_24	543
CUST_13	534
CUST_08	533
CUST_09	519
CUST_15	519
CUST_16	515
CUST_23	513
CUST_01	507
CUST_18	507
CUST_20	507
CUST_11	503
CUST_22	501
CUST_10	501
CUST_21	498
CUST_12	498
CUST_07	491
CUST_02	488
CUST_17	487
CUST_19	487
CUST_14	484
CUST_06	481
CUST_25	476
CUST_04	474
CUST_03	465

10. **Revenue by Discount Applied:** Generated Total Revenue by Discount Applied.

Revenue	discount_applied
1070892.5	No
545859.5	Yes

11. **Discount Applied by Location and Orders:** Compared to the Orders (Transaction ID) by Discount Applied and Location to check how many orders got placed by location and Orders.

discount_applied	location	orders
No	Online	4205
No	In-store	4151
Yes	Online	2149
Yes	In-store	2070

12. **Discount Applied by Sales:** Generated Total Sales by Discount Applied to see how many Sales generated based on the Discount Application.

discount_applied	sales
No	46413
Yes	23415

13. **Discount Applied by Transaction ID:** Frequent Purchase Compared to Discount Applied.

discount_applied	frequent_purchase
No	8356
Yes	4219

14. **Average Spending of Customers by all Purchasing factors:** Generated the Average of Total Spent by Location, Discount Applied, Total Revenue, Orders to understand the Purchasing behaviors of customers.

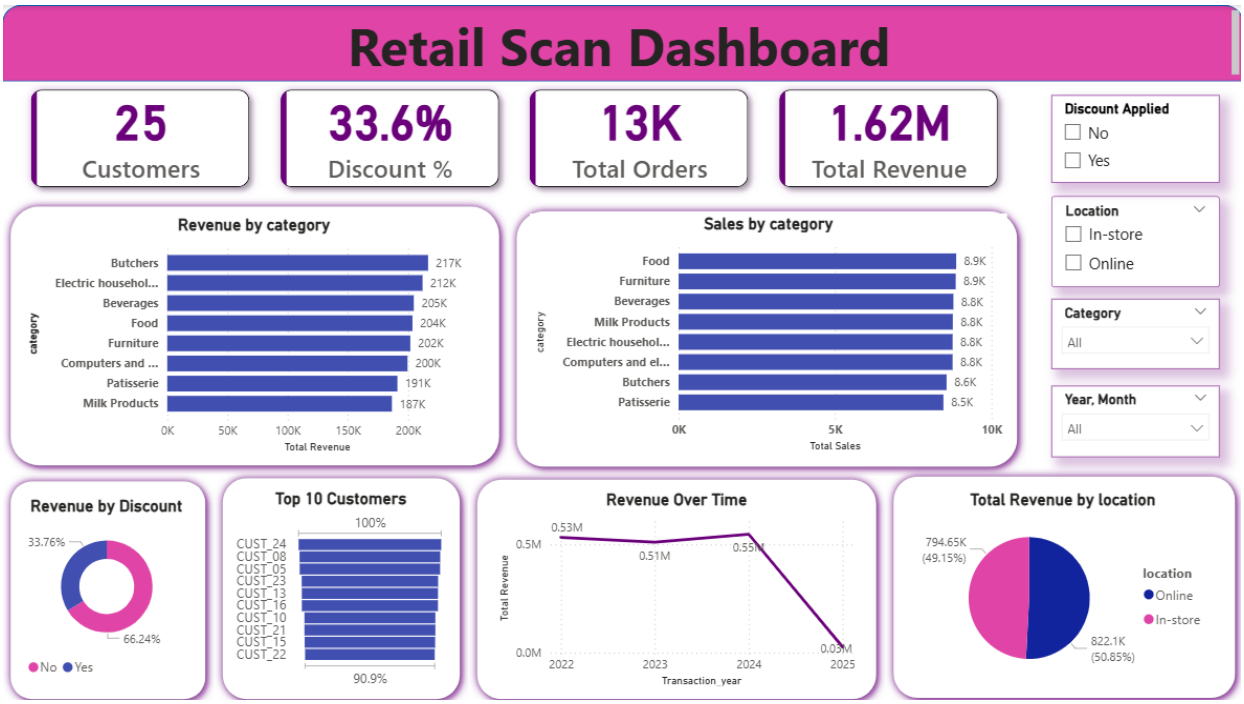
location	discount_applied	revenue	orders	Avg_spent
Online	Yes	277899	2149	129.31549557933923
Online	No	544202.75	4205	129.41801426872772
In-store	Yes	267960.5	2070	129.44951690821256
In-store	No	526689.75	4151	126.88261864610936

15. **Discount Applied usage Changes Over time:** Compared Discount applied by transaction years and Orders to see how the usage of Discount has evolved.

	discount_applied	Years	orders
▶	No	2024	2812
	No	2022	2731
	No	2023	2660
	Yes	2024	1429
	Yes	2022	1403
	Yes	2023	1327
	No	2025	153
	Yes	2025	60

5. Dashboard on Power BI

Finally, Built an interactive dashboard in **Power BI** to present insights visually.



6. Business Recommendations

- **Invest in Top Categories** - Prioritize Butchers, Electronics, and Beverages to maximize revenue growth.

- **Revive Low Performing Categories** - Bundle Milk Products and Patisserie with top sellers to boost sales.

- **Review Discount Policy** - Since non-discounted orders drive 66.2% of revenue, reduce broad discounts and apply them strategically.

- **Balance Sales Channels** - Maintain equal investment in Online and In-store as both channels perform equally well.

- **Channel Specific Marketing** - Promote Electronics Online and Furniture & Food In-store based on channel performance

- **Customer Loyalty Program** - Reward top spending and frequently purchasing customers to encourage repeat purchases.

- **Replicate 2024 Strategies** - 2024 recorded highest revenue, analyze and replicate those strategies for 2025 growth.

- **Improve Data Quality** - Implement better data collection at point of transaction to reduce missing values in future analysis.